

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code.	Yes	
2	Proper File Structure	Files and folders are logically organized	Yes	
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	
4	Function / Method Names	Functions are descriptively named	Yes	
5	Code Comments	Inline and block comments explain logic	Yes	
6	Modular Design	Code is split into reusable functions/modules	Yes	
7	No Redundant Code	Duplicate or unused code is removed	Yes	
8	Error Handling	Exceptions and errors are handled gracefully	Yes	

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Authentication Module	Python, Flask/Firebase	Login & Registration	High

2	Emotion Detection Module	Python, OpenCV, TensorFlow	Emotion Recognition	High
3	Learning Recommendation Engine`	Python, Generative AI	Personalized Learning Suggestions	High
4	Dashboard Module	HTML, CSS, JavaScript	User Progress & Analytics	Medium
5	Database Module	Firebase / MySQL	User Data & Emotion History	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure and modular implementation
Readability	5	Easy-to-understand code with meaningful names and comments
Reusability	5	Reusable modules reduce duplication and simplify maintenance
Documentation / Comments	5	Sufficient comments and documentation for future development
Overall Score	5/5	The project follows good coding standards, modular architecture, and maintainable design practices